QBS 103 Final Project

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Methods

Data Source and Preparation

This project used gene expression and maternal characteristic data from the Vitamin D Antenatal Asthma Reduction Trial (VDAART), described by Mirzakhani et al. (2016). The dataset (GSE85307) included 157 maternal blood samples: 110 from normal pregnancies and 47 from pregnancies with preeclampsia.

The gene expression data were matched to maternal characteristics using GEO sample IDs. Text prefixes were removed from the covariate values, and continuous variables were converted to numeric format. Maternal BMI was divided into two categories using a cutoff of 25 kg/m^2. All 157 samples were retained in the analysis.

Data Analysis and Visualization

Maternal age, gestational age at delivery, and maternal vitamin D levels were summarized using means and standard deviations. Pregnancy outcome and BMI category were summarized using counts and percentages.

Gene expression was visualized using histograms, scatterplots, and boxplots for three selected genes: *PTX3*, *ABCA9*, and *ACTG2*. Scatterplots included fitted linear trends, while boxplots displayed individual observations.

A heatmap was created using ten genes, with expression values standardized within each gene. Hierarchical clustering was applied to genes and samples, and annotation bars indicated pregnancy outcome and maternal BMI category. An alluvial plot was also used to visualize the distribution of samples across the two categorical variables.

The analysis was descriptive, and no formal hypothesis tests were performed.

Software

All data cleaning, analysis, and visualization were conducted using R version 4.5.1. Data processing and visualization used tidyverse (version 2.0.0), ggplot2 (version 4.0.2), and ggpubr (version 1.0.0). Heatmaps were generated using pheatmap (version 1.0.13), and alluvial plots were created using ggalluvial (version 0.12.6). The report was generated from R Markdown using knitr (version 1.52).

The R Markdown code and project files are available in the GitHub repository:https://github.com/zhualtis-gif/HSE103_FinalProject.

Results

The dataset included 157 samples, with 110 normal pregnancies and 47 preeclampsia pregnancies. Maternal age was similar between the two groups. Compared with the normal pregnancy group, the preeclampsia

Table 1: Maternal characteristics by pregnancy outcome. Continuous variables are presented as mean (SD), and categorical variables as counts and percentages.

Characteristic	Normal (n = 110)	Preeclampsia (n = 47)
Maternal age (years)	26.6 (5.1)	25.8 (4.9)
Gestational age at delivery (weeks)	39.3 (1.0)	37.8 (3.5)
Maternal vitamin D (ng/mL)	24.4 (14.5)	19.7 (8.3)
Maternal BMI below 25	40 (36.4%)	13 (27.7%)
Maternal BMI 25 or higher	70 (63.6%)	34 (72.3%)

group had a lower mean gestational age at delivery (37.8 vs. 39.3 weeks) and a lower mean maternal vitamin D level (19.7 vs. 24.4 ng/mL). The proportion of mothers with BMI >=25 was also higher in the preeclampsia group (72.3% vs. 63.6%) (Table 1).

Summary Statistics

Gene Expression Distributions

The expression distributions differed across the three selected genes. *PTX3* showed a slightly right-skewed distribution, while *ABCA9* expression values were more concentrated. Most *ACTG2* expression values were within a narrow range, with a few notably high observations (Figure 1).

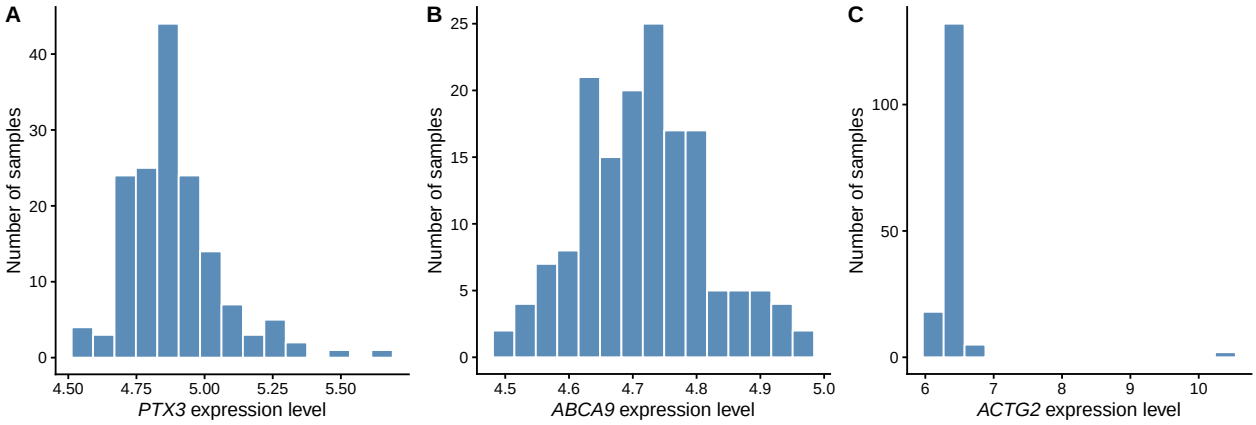


Figure 1: Gene expression distributions across 157 samples: (A) *PTX3*, (B) *ABCA9*, and (C) *ACTG2*. Each histogram shows the number of samples within each expression interval.

Maternal Age and Gene Expression

The scatterplots showed weak linear trends between maternal age and expression levels for all three genes. No clear linear relationship was visible for *PTX3*, *ABCA9*, or *ACTG2* (Figure 2).

Gene Expression by Pregnancy Outcome and BMI

Gene expression distributions overlapped across pregnancy outcome and BMI groups. Some differences in group medians were visible for *ABCA9*, while *PTX3* and *ACTG2* showed relatively similar expression levels across the groups (Figure 3).

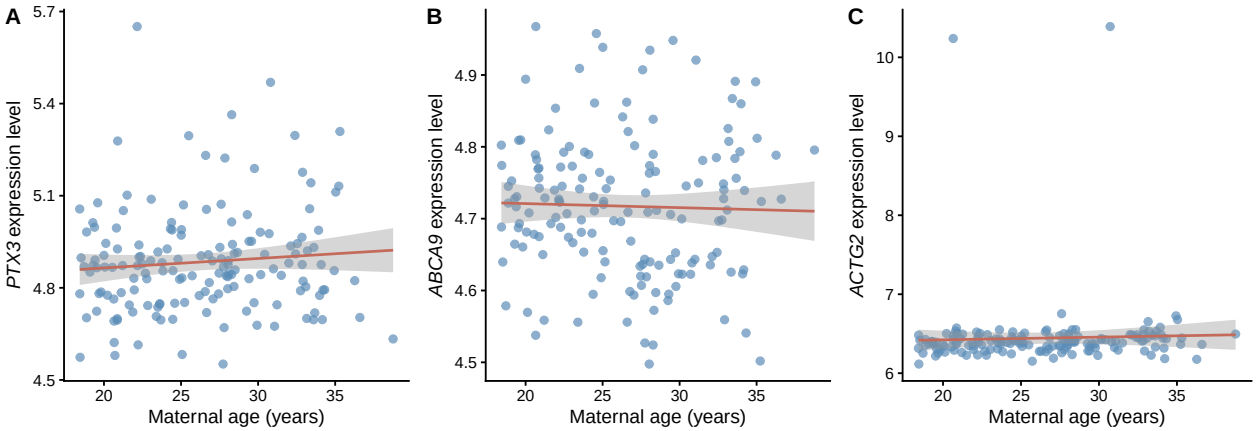


Figure 2: Gene expression levels in relation to maternal age: (A) *PTX3*, (B) *ABCA9*, and (C) *ACTG2*. Each point represents one sample. The fitted lines show linear trends, with shaded areas indicating 95% confidence intervals.

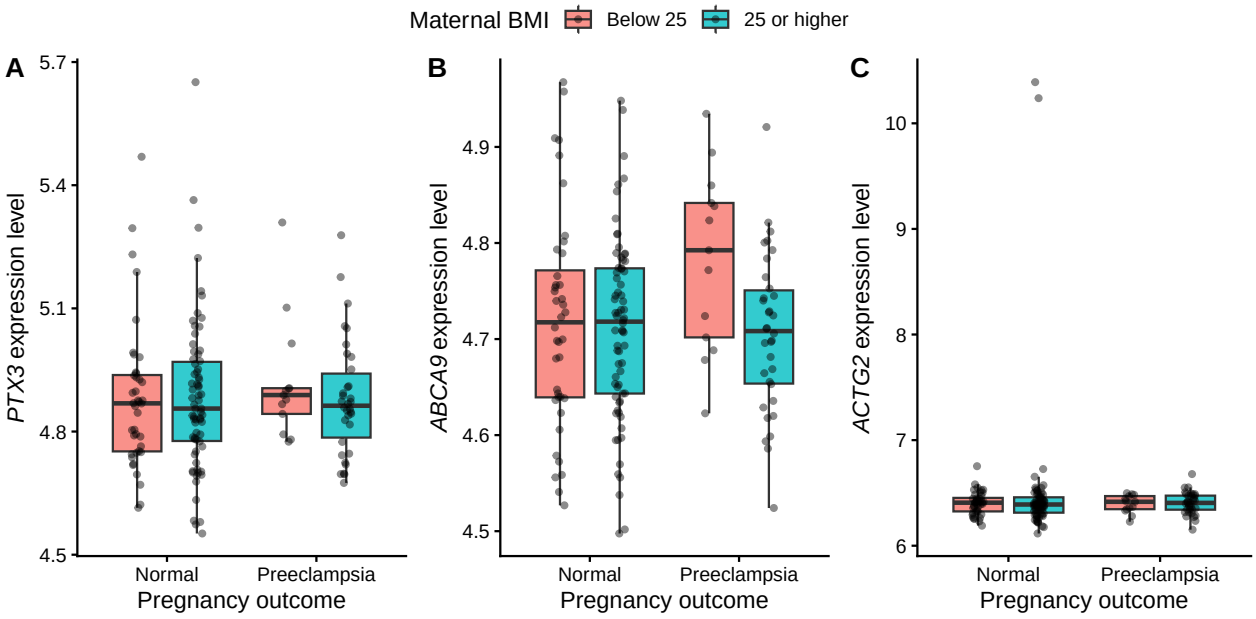


Figure 3: Gene expression by pregnancy outcome and maternal BMI category: (A) *PTX3*, (B) *ABCA9*, and (C) *ACTG2*. Boxes show the median and interquartile range, while jittered points represent individual samples.

Gene Expression Heatmap

Figure 4 shows the expression patterns of ten selected genes across 157 samples. Hierarchical clustering was applied to both genes and samples, with pregnancy outcome and maternal BMI category displayed as annotation bars. Expression levels varied across samples, with a few notably high values. The clustering did not show a clear separation between normal and preeclampsia samples.

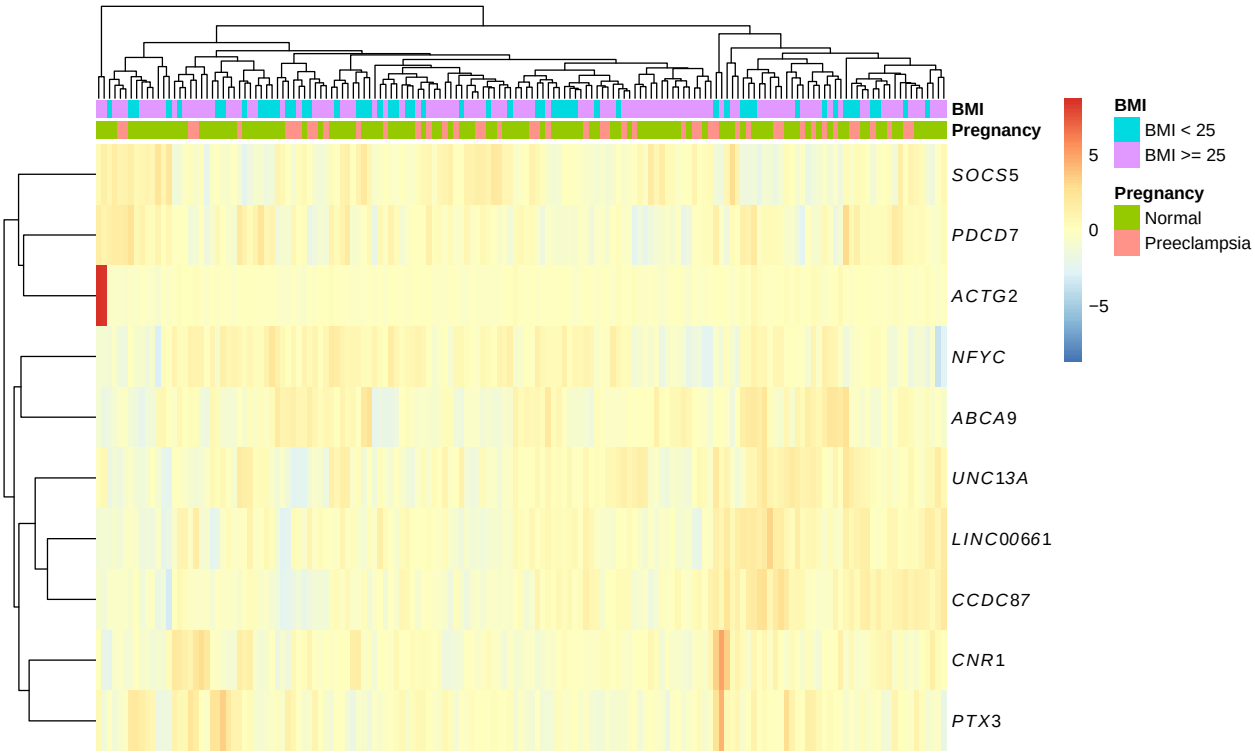


Figure 4: Heatmap of ten selected genes across 157 samples. Expression values were standardized within each gene. Rows and columns were clustered hierarchically. The tracking bars indicate pregnancy outcome and maternal BMI category.

Maternal BMI and Pregnancy Outcome

Figure 5 illustrates the distribution of maternal BMI categories across pregnancy outcomes. Among 157 samples, 53 mothers had a BMI below 25 and 104 had a BMI of 25 or higher. The flows show the number of normal and preeclampsia pregnancies within each BMI category.

AI Use Disclosure

I used ChatGPT (OpenAI) as a supplementary resource to discuss the project approach, troubleshoot R code, receive guidance on GitHub setup and submission procedures, and improve the clarity and wording of written explanations. I reviewed the code, analyses, visualizations, and interpretations and take responsibility for the submitted work.

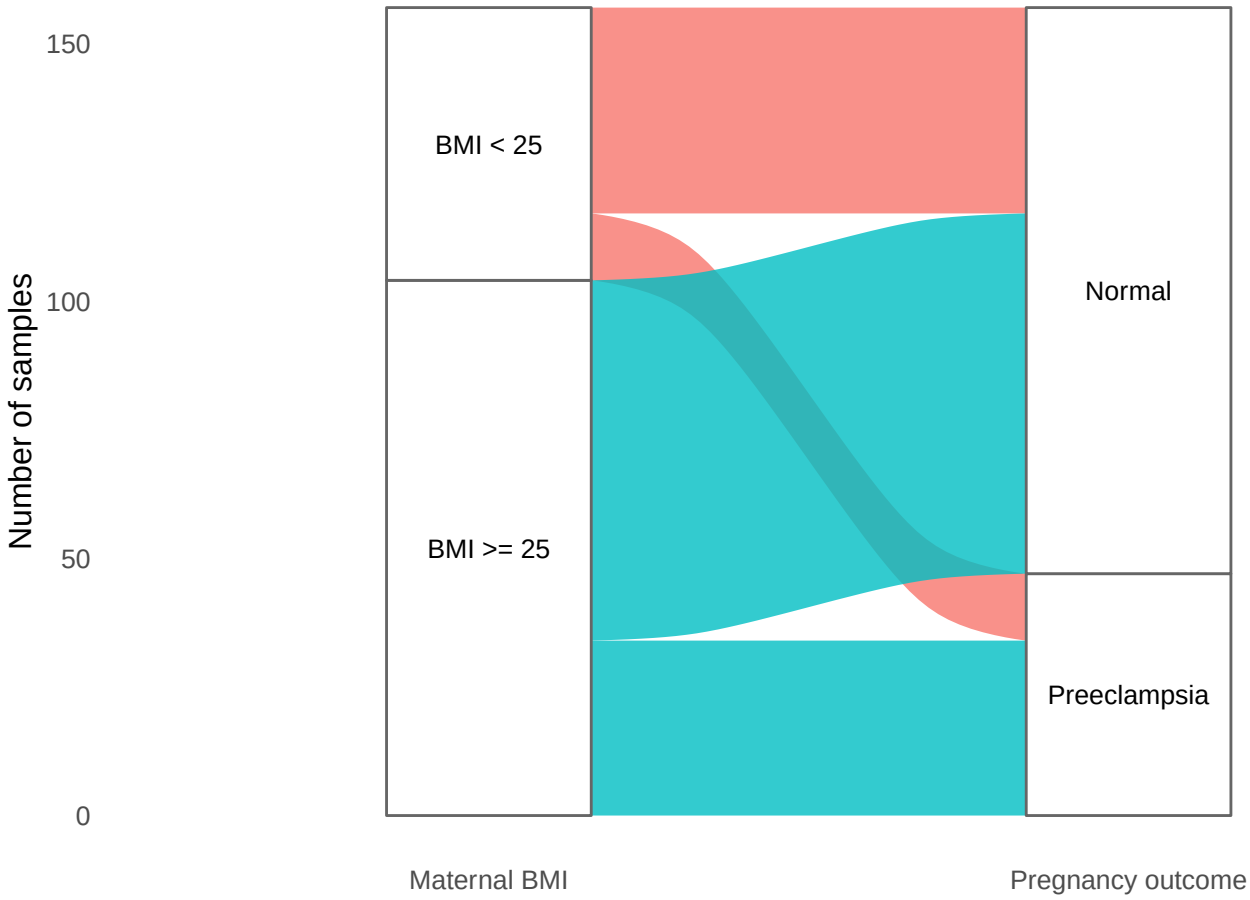


Figure 5: Distribution of maternal BMI categories across normal and preeclampsia pregnancies. The width of each flow represents the number of samples in that group.